Assignment 2 - LLM Document Analysis

CAP5619 | November 4, 2026 | Methodology Report

GitHub Repository: https://github.com/danielp382/CAP5619-Assignment2-LLM-Document-Analysis

Project goal

For this assignment, I used Python and a local large language model to review SEC Form 8-K documents and look for announcements about newly launched products. The output has five columns: company_name, stock_name, filing_time, new_product, and product_description. I limited product descriptions to fewer than 180 characters.

Tools and LLM setup

I worked in Google Colab for my early tests and then moved the Python scripts to Windows. I used Python 3.13, Ollama, and the local llama3.2:3b model (LLaMA 3.2, 3 billion parameters). I also used BeautifulSoup to clean HTML text from SEC filing exhibits. My Windows script calls Ollama through Python subprocess.run(), so the model runs locally rather than through a paid API. For SEC filing information, I used the public submissions API at data.sec.gov.

How I processed the filings

First, I downloaded SEC filing text and removed HTML formatting so the model could read the announcement. I sent the cleaned text to LLaMA with instructions to answer Yes or No for a new-product launch, return the exact product name, provide a short description, and explain its decision. The prompt also tells the model not to confuse earnings reports, financial changes, or ordinary company updates with new-product launches.

Next, I used Python to read the response line by line and save the values in a dictionary. I checked whether the description had fewer than 180 characters and whether the product fields were filled in. I retrieved the company name, stock ticker, and acceptance timestamp by matching the filing accession number in the SEC JSON data. Finally, I used the csv library to write the required five-column CSV and used a separate script to confirm the column names and description length. The model performs the product-name/entity extraction; the SEC API provides the official company and time information.

Documents tested and results

I manually tested two SEC documents from two companies: TELA Bio, Inc. (Form 8-K exhibit dated August 16, 2023, accession 0001104659-23-092205) and Microsoft Corp. (Form 8-K dated October 28, 2026, accession 0001193125-26-436149). TELA Bio announced OviTex PRS Long-Term Resorbable, so the model correctly returned Yes and a product description under 180 characters. The saved example description was 143 characters. Microsoft discussed segment reporting and recast financial statements, so the model correctly returned No and N/A for the product fields.

In this tested sample, I processed 2 documents from 2 companies and produced 1 valid product-announcement CSV row. Both classifications matched my manual review of the documents. I did not reach 100 valid product entries in this sample, so the current CSV is a partial dataset rather than a complete 100-entry collection.

Challenges and limitations

One issue was that not every 8-K announces a product. Some filings only contain financial or business information. I also had to find the correct exhibit and make sure I used the right accession number, because the most recent filing for a company can change. Another issue was Windows text encoding when reading Ollama output. I fixed that by setting UTF-8 encoding in subprocess.run().

The local model sometimes changes the wording of a description between runs, even when the product name stays the same. My Python checks can identify missing values and long descriptions, but they cannot prove that every model statement is factually correct. I manually checked the two test examples against their SEC documents. The current implementation extracts one product per model response, so a filing that announces multiple products may need additional processing and manual review. A larger dataset would require processing more filings and checking the positive results.

Sources and code

SEC EDGAR filings and exhibits: https://www.sec.gov/Archives/edgar/ ; SEC submissions API: https://data.sec.gov/submissions/ . The Python scripts, sample CSV, and processing notes are included in my GitHub repository listed at the top of this report.